

# Aryan Kumar Srivastava

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## Summary

Backend-focused Computer Science undergraduate building networked systems and developer tools in Go, TypeScript, and Node.js. Open-source Node.js contributor with hands-on work in TCP/IP, concurrency, HTTP/1.1, Redis protocols, performance testing, and full-stack web development.

## Open Source Experience

### Node.js (nodejs/node) – Open Source Contributor

Jun 2026–Aug 2026

*Core runtime contribution*

*Remote*

- Fixed an **EventTarget** weak-listener retention bug by replacing overwritten listener entries with **SafeSet** collections, weak retainer references, and matching cleanup logic.
- Added two regression tests covering garbage collection and independent listener removal; incorporated maintainer feedback through iterative review to merge PR #64024.

## Projects

### Clockwork – Redis-compatible key-value store (GitHub)

*Go*

- Built a Redis-compatible TCP key-value store using only Go's standard library; implemented RESP so unmodified **redis-cli** and **redis-benchmark** clients work directly.
- Implemented string and list commands, TTL expiration, and atomic SAVE persistence; coordinated concurrent clients with goroutines and **sync.RWMutex**.
- Benchmarked 100,000 requests with 50 concurrent clients; optimized amortized O(1) LPUSH and released a Docker image plus cross-platform binaries.

### servee – HTTP/1.1 server from raw TCP sockets (GitHub)

*TypeScript, Bun*

- Developed a zero-dependency HTTP/1.1 server over raw TCP sockets with request parsing, routing, response encoding, body streaming, and persistent connections.
- Recorded 29,347 requests/sec at 2.96 ms average latency in a local Autocannon benchmark; Express 5.2.1 recorded 16,306 requests/sec at 5.51 ms under the same setup.
- Validated protocol and connection behavior with unit and integration tests.

### Feedback – HTTP load-testing CLI (GitHub)

*Go*

- Created a concurrent HTTP load-testing CLI with configurable workers and round-robin distribution across multiple endpoints.
- Instrumented Time to First Byte with **httptrace**, reported request failures and timing statistics, and published cross-platform binaries through GitHub Releases.

### Eiga – Viewing-history visualizations (Live)

*Next.js, TypeScript, D3.js*

- Built a privacy-first browser application that transforms Letterboxd exports into interactive visualizations across watch date, rating, decade, and two additional dimensions.
- Kept imported data entirely in the browser without accounts or server-side storage; deployed on Cloudflare Pages.

## Technical Skills

**Languages:** Go, TypeScript, JavaScript, C, SQL, PL/SQL, Bash

**Backend:** Node.js, Bun, Fastify, REST APIs

**Systems:** TCP/IP, HTTP/1.1, RESP, networking, concurrency

**Databases:** Redis, MongoDB, SQL, DBMS

**Frontend:** React, Next.js, D3.js

**Cloud/Tools:** AWS, Docker, Git, Linux, Cloudflare Pages, Vercel, Fly.io

**Testing/Performance:** Unit and integration testing, Autocannon, **httptrace**, load testing, benchmarking

## Education

**KIET Deemed to be University**

2024–2028

*B.Tech in Computer Science and Engineering, specialization in Artificial Intelligence*

## Certifications

**AWS Certified Cloud Practitioner** (Dec 2025–Dec 2028) | **Cisco Introduction to Cybersecurity** (Jul 2025) | **Cisco Junior Cybersecurity Analyst Career Path** (Jun 2026)